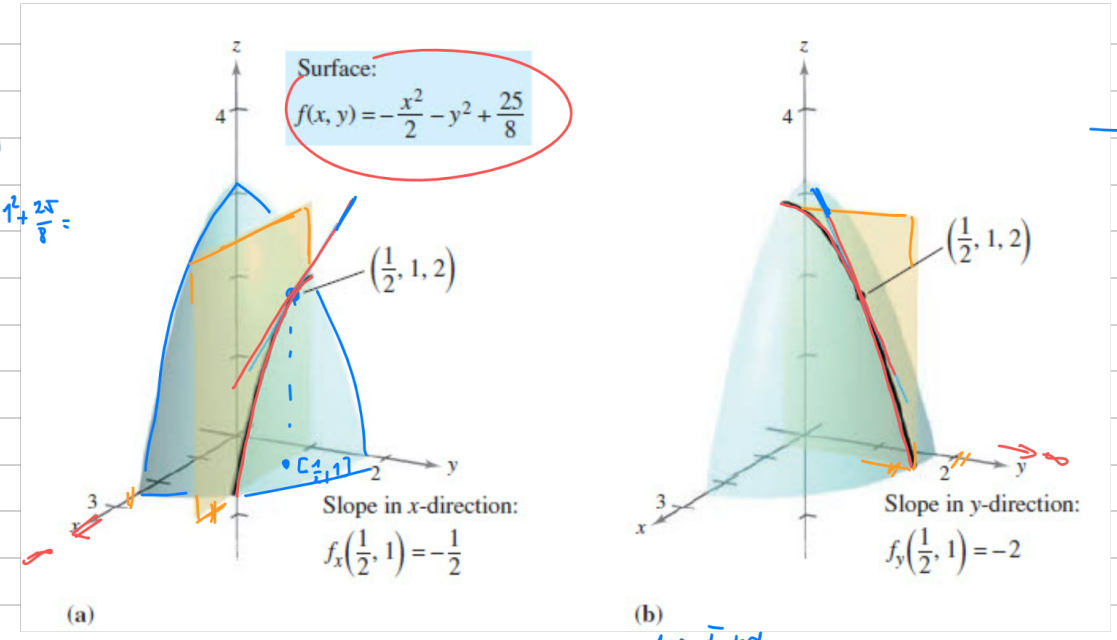
PARCIÁLNE DERIVÁCIE

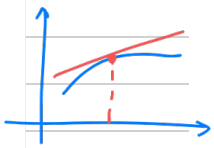
PRÍKLAD $f(x,y) = -\frac{x^2}{2} - y^2 + \frac{25}{8}$ nájdite deriváciu v bode $(\frac{1}{2}, 1, 2)$

pozn.
BOD $= [\frac{1}{2}, 1]$

$f(\frac{1}{2}, 1) = -\frac{\frac{1}{4}}{2} - 1^2 + \frac{25}{8} =$

$= 2$





dosadí bod

$f_x(x,y) = \frac{\partial f(x,y)}{\partial x} = -\frac{2x}{2} + 0 + 0 = -x$

$f_x(\frac{1}{2}, 1) = -\frac{1}{2}$

$f_y(x,y) = \frac{\partial f(x,y)}{\partial y} = 0 - 2y + 0 = -2y$

$f_y(\frac{1}{2}, 1) = -2 \cdot 1 = -2$

DERIVÁCIE VYŠŠÍCH RÁDOV

1D

$f'(x)$ – smernica dotyčnice

$f''(x)$ – konvexná, konkávna

2D

$\frac{\partial f}{\partial x}$ $\frac{\partial f}{\partial y}$

$f_{xx} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2}$ $\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y} = f_{xy}$

$f_{yx} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x}$ $\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2} = f_{yy}$

4 možné parciálne derivácie

PRÍKLAD $f(x,y) = 3xy^2 - 2y + 5x^2y^2$

Ak $f(x,y)$ je spojitá $f_{xy} = f_{yx}$

1. derivácie

$f_x = 3y^2 - 0 + 10xy^2$

$f_y = 6xy - 2 + 10x^2y$

2. derivácia

$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = 6y + 20xy$ $\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = 0 + 10y^2$

$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = 6y + 20xy$ $\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = 6x + 10x^2$

Výpočty $f_{xx}(-1,2) = 0 + 10y^2\Big|_{(-1,2)} = 10\,2^2 = 40$

$f_{yx}(-1,2) = 6\cdot 2 + 20\cdot(-1)\,2 = 12 - 40 = -28$

DERIVÁCIA V BODE = ČÍSLO

DERIVÁCIA = FUNKCIA

3D $f(x,y,z)$

$\frac{\partial f}{\partial x}$ $\frac{\partial f}{\partial y}$ $\frac{\partial f}{\partial z}$ — 3x 1. parciálna derivácia

$\frac{\partial^2 f}{\partial x^2}$ $\frac{\partial^2 f}{\partial x \partial y}$ $\frac{\partial^2 f}{\partial x \partial z}$

$\frac{\partial^2 f}{\partial y \partial x}$ - -

$\frac{\partial^2 f}{\partial z \partial x}$ - - $\frac{\partial^2 f}{\partial z^2}$

9x 2. parciálna derivácia

(TOTÁLNY) DIFERENCIÁL (total differencial)

Definícia: $z = f(x,y)$ potom totálny diferenciál je funkcia v premennej z

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$$

iný spôsob zápisu:

$$dz = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$$

PRÍKLAD: $z = 2x\sin y - 3x^2y^2$

totálny diferenciál:

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$$

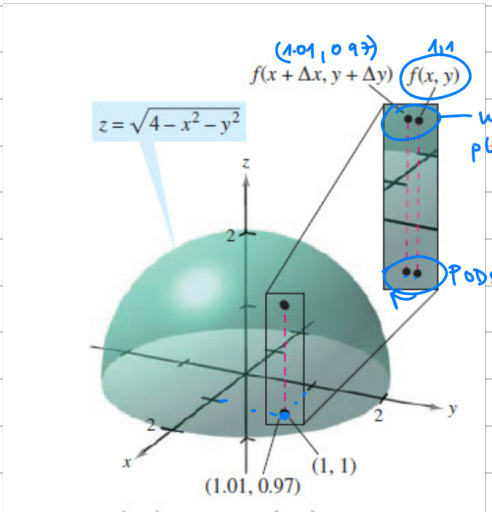
$$= (2\sin y - 6xy^2)\,dx + (2x\cos y - 6x^2y)\,dy$$

?? NAČO JE TO DOBRÉ ??

Používajú to počítače na výpočet hodnoty funkcie

PRÍKLAD:

$z = \sqrt{4-x^2-y^2}$ chcem popísať pohyb z bodu $(1,1)$ do bodu $(1.01, 0.97)$



As (x, y) moves from $(1, 1)$ to the point $(1.01, 0.97)$, the value of $f(x, y)$ changes by about 0.0137.

$(x,y) = (1,1)$ (x, y)

$(x+\Delta x, y+\Delta y) = (1.01, 0.97) =$

$= (1 + 0.01, 1 - 0.03)$ (+0.01, −0.03)

x Δx y Δy

$\Delta z = dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$ to isté

$\frac{\partial z}{\partial x} = \left[(\quad)^{\frac{1}{2}}\right]'_x = \frac{1}{2}\cdot(4-x^2-y^2)^{-\frac{1}{2}}\cdot(0-2x+0)$

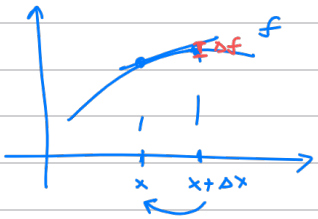
$= \frac{-x}{\sqrt{4-x^2-y^2}}$

$\frac{\partial z}{\partial y} = \frac{-y}{\sqrt{4-x^2-y^2}}$

$$\frac{\partial z}{\partial x}(1,1) = \frac{-1}{\sqrt{4-1^2-1^2}} = \frac{-1}{\sqrt{2}}$$

$$\frac{\partial z}{\partial y}(1,1) = -\frac{1}{\sqrt{2}}$$

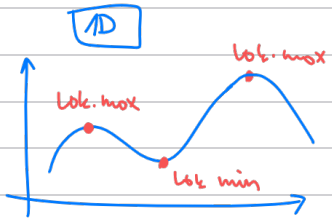
$$dz = -\frac{1}{\sqrt{2}} \cdot 0{,}01 + -\frac{1}{\sqrt{2}} \cdot (-0{,}03) \approx 0{,}0141$$

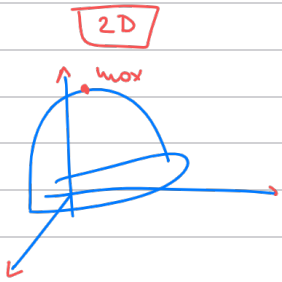


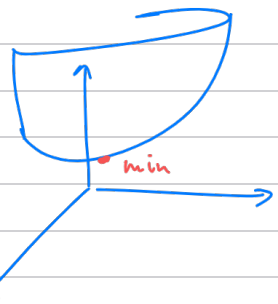
$$\lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = f'(x)$$

DIFFERENCE $\Delta z = f(1.01, 0.97) - f(1,1)$

IDEME HĽADAŤ LOKÁLNE EXTRÉMY







PRÍKLAD $f(x,y) = 1 + 6y - y^2 - xy - x^2$

NÁJDI lokálne extrémy.

GEOGEBRA 3D

WOLFRAM ALPHA

① Kandidáti = stacionárne body

$$\frac{\partial f}{\partial x} = 0$$

$$\frac{\partial f}{\partial y} = 0$$

$$\frac{\partial f}{\partial x} = 0 + 0 + 0 - y - 2x \Rightarrow$$

$$\frac{\partial f}{\partial y} = 0 + 6 - 2y - x + 0$$

1D

$f'(x) = 0$

$$\begin{cases} -y - 2x = 0 \\ 6 - 2y - x = 0 \end{cases}$$

↓

Vyjadríme

$-y - 2x = 0 \rightarrow y = -2x$

$6 - 2y - x = 0$

$6 - 2.(-2x) - x = 0$

$6 + 4x - x = 0$

$6 + 3x = 0$

$3x = -6$

$x = -2$

$y = -2x$

$y = -2 \cdot -2 = 4$

Stacionárny bod $A = [-2, 4]$ ← kandidát na extrém

1D

$f'(x) = 0$

$f'' > 0$ min

$f'' < 0$ max

2D

$f''_{xx} = ?$

$$|H_f| = \begin{vmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{vmatrix}$$

$H_f > 0$ → je tam extrém $f''_{xx} < 0$ max
$f''_{xx} > 0$ min

$H_f < 0$ → nemá extrém

$H_f = 0$ → nevieme povedať (sedlový bod alebo ekvivalent)

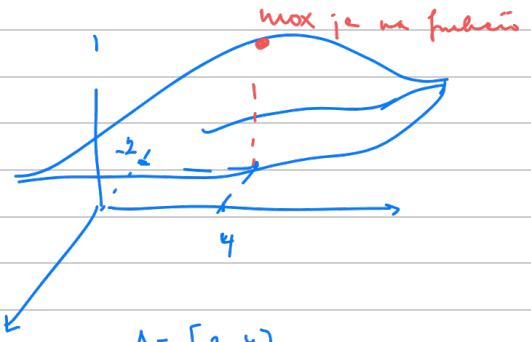
$f'_x = -y - 2x$
$f'_y = 6 - 2y - x$

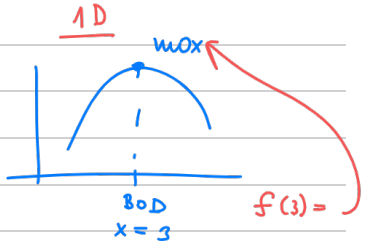
$\frac{\partial^2 f}{\partial x^2} = -2$ $\quad \frac{\partial^2 f}{\partial x \partial y} = -1$

$\frac{\partial^2 f}{\partial x \partial y} = -1$ $\quad \frac{\partial^2 f}{\partial y^2} = -2$

$$H_f = \begin{vmatrix} -2 & -1 \\ -1 & -2 \end{vmatrix} = \underbrace{-2 \cdot -2}_{4} - \underbrace{(-1 \cdot -1)}_{1} = 4 - 1 = 3 > 0$$ — JE TAM EXTRÉM

$f''_{xx} = -2 < 0 \longrightarrow$ JE TAM MAXIMUM





maximum je vždy na hranici

$A = [-2, 4]$

$f(A) = 1 + 6 \cdot 4 - 4^2 - (-2) \cdot 4 - (-2)^2$

$= 13$

ODPOVEĎ: funkcia f má lokálne maximum v bode $A = [-2, 4]$ s hodnotou 13 $\quad f(A) = 13$

TEÓRIA: ak $f(x,y)$ má stacionárny bod A a $f(x,y)$ je 2x diferencovateľná v tomto bode

tak platí

① A stacionárny bod $\Leftrightarrow \frac{\partial f}{\partial x}(A) = 0 \quad \wedge \quad \frac{\partial f}{\partial y}(A) = 0$

② $f(x,y)$ má v bode A extrém a platia tvrdenia

$H_f(A) > 0$ $\quad \frac{\partial^2 f}{\partial x^2}(A) > 0$ minimum

$\frac{\partial^2 f}{\partial x^2}(A) < 0$ maximum

$H_f(A) = 0$ → nevieme určiť

$H_f(A) < 0$ → nemá extrém